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Democracy and Militarization in Developing Countries: A Panel Vector Autoregressive Analysis

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ABSTRACT

This study examines the dynamic relationship between democracy and the military in more than 40 developing countries from 1990 to 2017. We investigate the dynamic interaction between democracy and military institutions using a panel vector autoregressive model and impulse response functions as well as variance decomposition analyses. We show that democracy plays a significant role in the substitution of nonmilitary expenditures for defense expenditures. We also investigate the response of democracy to positive shocks in military and nonmilitary expenditures. We find that the responses of political systems and different indexes of democracy including electoral, liberal, participatory, deliberative, and egalitarian democracy to positive shocks in military expenditures are negative and significant, whereas their responses to the shocks in nonmilitary expenditures are not significant. This result suggests that the political behavior of governments in developing countries is influenced more heavily by their spending on the military sector than by their spending on the nonmilitary sector.

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Introduction

The increase in military expenditures worldwide is a major burden for economies of developing countries. The latest Stockholm International Peace Research Institute yearbook (SIPRI 2020) shows that world military spending was US$ 1917 billion (2.2% of world GDP or US$ 250 per person): The level of military expenditures in 2019 was 3.6% higher than in 2018 and 7.2% higher than in 2010. According to the SIPRI report, the growth of global military expenditures in 2019 was 'the fifth consecutive annual increase and the largest of the decade 2010–19, surpassing the 2.6 per cent rise in 2018.' On the other side, the annual reports of Varieties of Democracy (V-Dem), which produces the largest global dataset on democracy for 202 countries from 1789 to 2018, show a worsening situation with reference to the quality of democratic institutions worldwide. Their 2019 report titled 'Democracy Facing Global Challenges' (V-Dem, 2019) mentions that 'almost one-third of the world's population lives in countries undergoing autocratization, surging from 415 million in 2016 to 2.3 billion in 2018.' The democracy report for 2020 titled 'Autocratization Surges – Resistance Grows' (V-DEM, 2020) shows a deterioration of democratic institutions. It shows that for the first time since 2001, autocracies are in the majority (92 countries – home to 54% of the global population). Their latest report, published in 2021, illustrates a further decline in quality of democratic institutions, especially in the global south. Their report (V-DEM, 2021), 'Autocratization Turns Viral,'

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shows that autocracies are home to 68% of the world population. It further shows that the worldwide decline during the past 10 years continues in 2020, especially in the Asia-Pacific region, Central Asia, Eastern Europe, and Latin America.

These conditions give rise to a natural question: Is dictatorship the result of militarism or the cause of it? Despite the significant body of literature that has examined how democracy impacts military systems, few studies have explored the political impact of military expenditures in developing countries. The differences among developing countries in the relative size of their military burdens result from their different economic and political conditions (Maizels and Nissanke 1986). A strand of literature argues that democracies spend less on the military than autocracies (Kimenyi and Mbaku 1996; Dunne and Perlo-Freeman 2003; Fordham and Walker 2005; Dizaji, Farzanegan, and Naghavi 2016). In a democratic state, internal conflicts are settled by compromise and in nonviolent ways, providing the conditions in which external conflicts between democratic states can also be settled peacefully. Autocratic regimes' survival depends largely on military power, while democracies pay more attention to nonmilitary sectors (Yildirim and Sezgin 2005; Dahlum and Knutsen 2017; Dizaji 2019a,b). Democracy may create a positive but indirect effect upon economic growth via the channel of liberty-to-innovation, conditioned by the level of economic development (Tang and Tang 2018).

While political institutions can affect military expenditures, they may also be affected by the militarization process. Military pressure impedes citizens' demands for democracy. Non-democratic regimes spend greatly on the military and security sectors, which blunts their populations' democratic aspirations (Ross 2008; Dizaji 2019b). Moreover, military expenditures may crowd out productive nonmilitary expenditures, such as education and health expenditures (Evans and Rose 2007), which are essential for establishing and promoting democracy, and slow economic growth (Rothschild 1973).

Political institutions and military expenditures are both endogenous and conventional methods that implicitly treat political institutions as exogenous variables may be biased because political institutions also respond to militarization. To the best of our knowledge, few studies have investigated the simultaneous relationships between political development and militarization. This study contributes to the literature by investigating the dynamic simultaneous interactions between militarization and democracy.

We design a panel vector autoregression (PVAR) model for 47 developing countries that focuses on the dynamic relationship between the military and political systems, enabling us to examine how democracy affects government budgets and military and nonmilitary spending and how military expenditures influence democracy indexes. Vector autoregression models can treat all variables as endogenous and reveal the dynamic interactions between them (Sims 1980; Enders 1996). This is an important advantage that has been neglected in most previous studies, which have applied static single-equation models. Moreover, while most of the studies on the military–democracy nexus have used a single measure of democracy (e.g. Polity index), we use a newly released democracy-measuring dataset from the Varieties of Democracy project, which categorizes democracy into electoral, liberal, deliberative participatory, and egalitarian components. This dataset allows us to investigate democratization using measures that are more reliable, valid, comprehensive, and nuanced than those used in previous studies (Bergougui and Murshed 2020).

In the next section, we review the relevant literature. In the third section, we describe the study's data and methodology. In the fourth section, we report the empirical results. The final section provides concluding remarks.

Literature Review

The interaction between political regimes and military expenditures has been a research issue for years. Studies have examined this nexus from different analytical and methodological perspectives. We review some of these studies and then explain the research design we use to shed more light on the dynamic relationship between political institutions and military expenditures.

The Effect of Political Institutions on Military Expenditures

Kotera and Okada (2017) address the question '*How does democratization affect the composition of government expenditure?*' using data drawn from a sample of 125 countries covering 1972 to 2010. Their results show that democratization does not have a significant impact on total expenditure but increases expenditure on health, and that an improved quality of political systems decreases military expenditures. The relationship between various democratic characteristics and states' fiscal capacity has been investigated by Rota (2016). The study shows that political participation is most effective in reducing military expenditures, while other elements of democracy play a distant secondary role. In a theoretical and empirical study, Dizaji, Farzanegan, and Naghavi (2016) examine the impact of political institutions on various categories of government expenditure in Iran. Their results show that improvements in political institutions have a decreasing impact on military expenditures and a positive impact on non-military expenditures such as education and health. In another study, Töngür and Elveren (2015) examine the relationships between military expenditure, income inequality, welfare regime type, and political regime. They show a significantly negative relationship between social democratic welfare regimes and military expenditures. The nexus between the type of political regime and military spending is also examined by Brauner (2015). Her results show that democracies spend less on its military than autocracies do, while an absolute dictatorship spends around 40% more on its military than a full democracy does. Elbadawi and Keefer (2014) examine the apparently low military expenditures in democracies and its implications for democratization. They conclude that the shift toward democracy in the Arab world (e.g. the 2011 Arab Spring) will not lead to a quick decline in military expenditures. The effects of form of government (presidential vs. parliamentary), electoral rules (plural vs. proportional), and concentration of parliamentary party groups on military expenditures are examined by Albalate, Bel, and Elias (2012). They conclude that democracies have lower percentages of military expenditure than autocracies and that presidential systems and majoritarian voting rules favor military expenditures. A number of other studies have also identified a negative effect of democratization on military spending (e.g. Lskavyan 2011; Yildirim and Sezgin 2005; Fordham and Walker 2005; Pereira 2004; and Kimenyi and Mbaku 1996).

Another line of literature examines how international sanctions on specific countries may force political institutions of receivers of sanctions to downsize their military expenditures. For example, the imposed sanctions by the United Nations, the European Union and United States on Iran have reduced an important source of funding of military projects (i.e. oil rents), resulting in significant decreases in its military spending (for more details see Dizaji 2014; Dizaji and Farzanegan 2021; Farzanegan 2011, 2014, 2021). Dizaji and Murshed (2020) investigate the impact of external arms embargoes on military expenditure and external and internal conflicts in developing countries in the post-Cold War period. The results of their PVAR models indicate that while arms restrictions do not significantly impact religious tensions and external conflicts, they may intensify ethnic tensions and internal conflicts in developing countries.

The Effect of Military Expenditures on Political Institutions

There are a few studies addressing the impact of militarization on the quality of democracy. Increases in the share of military expenditures could be associated with more secrecy and lower transparency (e.g. for national security reasons), which also weaken the quality of democratic institutions. Gupta, de Mello, and Sharan (2001) show that higher military expenditures is positively associated with public corruption.[1] Thus, military spending may also degrade the democratic quality of institutions through its increasing effects on corruption. Some studies such as Henderson and Singer (2000) show that an increase in the share of military spending crowds out the share of social and health spending and leads to frustration of the population and violence in the long-term. Such negative effects of military expenditures may then degrade the quality of political institutions. Collier and

Hoeffler (2007) also explain that military spending is not an effective instrument to reduce internal conflict. Countries with a larger share of military spending may provide necessary resources for potential opposition groups within the military to organize coups and strengthen their political power, further weakening the quality of political institutions. Using the panel data covering the 1984–2012 period for more than 100 countries, Farzanegan and Witthuhn (2017) find a robust and negative effect of military expenditures on the stability of a political system. Drivers of the wide differences in democracy levels across Latin America during the Cold War were examined by Bowman (1996). His estimations show that (a) military expenditures have a significantly negative effect on democracy scores over time and (b) military participation ratios have a more robust negative impact.

Summary of Literature and Our Contribution

Overall, the literature suggests that democracy and military expenditures go hand in hand. However, while the majority of studies have focused on the effect of political institutions in the allocation of government spending to military and non-military categories, there is a lack of empirical studies focusing on possible two-way relationships between political institutions and government spending structures. Our proposed PVAR model simulates the response of political institutions to a positive shock in military and non-military expenditures while also capturing the simultaneous impact of political institutions on military and non-military expenditures. The results extend the literature on the determinants of democracy by focusing on the role of the composition of government expenditures and provide useful insight on the possible response of political institutions to unexpected changes in different types of government expenditures. Our methodological approach is related to the study conducted by Dizaji, Farzanegan, and Naghavi (2016) which explored the response of military and non-military expenditures of the Iranian government to shocks in the quality of political institutions. We extend it by raising a new question on the effects of shocks in government spending on institutions and by expanding the analysis to larger number of countries in the global south.

Research Design

Data Description

We consider a panel of annual data from 47 developing countries covering 1990 to 2017 according to data availability. The list of countries is presented in the appendix. These selected countries provide an appropriate sample for generalizing our empirical findings and therefore, the likelihood of sample selection bias is small. First, the selected sample represents 69 percent of the world's population and 59 percent of global armed forces personnel in 2017. Moreover, according to SIPRI (2020), these countries have formed 67 percent of worldwide arms imports in 2017. Second, the issue of military burden and military in politics is a more common observation with the global south countries than the developed ones with established democratic institutions. Third, the geographical coverage of these 47 countries shows inclusion of different global regions such as Asia, Africa, South and Central America and Eastern Europe.

We consider the following variables to analyze the dynamic interrelations between military factors (military expenditures, arms imports, armed forces), government budget, and democracy: the arms imports to GDP ratio (*armimp*), military expenditures to GDP ratio (*milex*), nonmilitary expenditures to GDP ratio (*non-milex*), armed forces personnel to total labor ratio (*armforce*), government revenue to GDP ratio (*govreven*), military in politics index (*militpolit*), and the Polity2 index (*polity*). The nonmilitary expenditures to GDP ratio is the difference between the total general government expenditures to GDP ratio and the military expenditures to GDP ratio. The data on military expenditures, arms imports, and armed forces are collected from the World Bank's *World Development Indicators* (WDI) online database (World Bank 2019). The data on total general

government expenditures and government revenues are taken from the International Monetary Fund's *World Economic Outlook Database* (October 2019).[2]

Data for the military in politics index (*militpolit*) are collected from the International Country Risk Guide (ICRG, 2019). The index ranges from 0 to 6; higher scores indicate lower degrees of military involvement in politics, and lower scores indicate higher levels.

We consider several democracy components, using indexes for electoral democracy (*elecdem*), liberal democracy (*liberdem*), deliberative democracy (*delibdem*), egalitarian democracy (*egalitdem*), and participatory democracy (*participdem*), as reported by the Varieties of Democracy (V-Dem) project, version 9. The V-Dem dataset captures annual changes in politics and the quality of various aspects of democracy. It captures a greater range of variation and has the advantage of considering a variety of sub-components that are helpful in capturing more aspects of democracy. All of these democracy indexes range between 0 and 1, with higher values indicating a higher quality of democracy. Table A in the appendix presents the summary statistics of our variables.

Our main hypotheses are as follows:

H1: An improvement in the quality of political institutions decreases government expenditures on military and increases non-military expenditures.

H2: Militarization has a negative impact on a government's non-military expenditures and political institutions.

Methodology

This study applies a panel vector autoregressive (PVAR) model to estimate the possible interrelationships among the variables. Using PVAR models is appropriate for this research because they enable the investigation of dynamic simultaneous effects and interactions between military expenditures, government budget, and political institutions. We can thus capture the long-run variations in political institutions, military factors, and government budgets over time influenced by military expenditures and democracy shocks using the VAR approach. The VAR methodology is helpful in the absence of a priori theory regarding the simultaneous relationships among military factors, political institutions, and government budgets because it treats all variables as if they were jointly endogenous and does not force any a priori restrictions on the structural relationships among them. The PVAR model merges the standard VAR method with the panel-data approach (Love and Zicchiano 2006). The VAR model defines the variables on the left-hand side of the system based on predetermined lagged variables (Filippaki and Mamatzakis 2009). The econometric reduced-form model is given as follows:

$$Z_{it} = \Gamma(L)Z_{it} + \mu_i + \varepsilon_{it} \tag{1}$$

where i denotes the country, $t = 1, \ldots T$, Z_{it} includes a vector of stationary variables, Γ(L) denotes a matrix polynomial in the lag operator with $\Gamma(L) = \Gamma_1 L^1 + \Gamma_2 L^2 + \ldots + \Gamma_P L^P$, μ_i is a vector of intercepts, and ε_{it} represents a vector of possibly contemporaneously correlated reduced-form disturbances (Dees and Guntner 2014). Since only lagged values of the endogenous variables appear on the right-hand side of the VAR model, simultaneity is not a problem, and the OLS estimations are consistent. In standard VAR models, the disturbance terms are generally characterized by simultaneous correlations, so that the reactions of the system to changes in a particular variable indicate the responses of all those variables that have simultaneous correlations with it. However, this simultaneous correlation is purified by the Cholesky orthogonalization mechanism. The dynamic changes of variables in response to shocks in a specific variable are illustrated by impulse response functions (IRFs), which enable us to investigate the dynamic impacts of shocks to a particular variable (e.g.

military expenditures) on the other variables in the model. The IRFs allow us to identify the magnitude and statistical significance of such responses to a one standard deviation increase in the shock variable (Stock and Watson, 2001). The whole system is examined by using variance decomposition analyses. Variance decomposition attributes the variance of forecast errors in a particular variable to its own shocks and the shocks to the other variables in the system (Brown and Yücel 1999).

Empirical Results

The ordering of the variables is essential in VAR models, as different orderings may result in different responses within a VAR system. To test our first hypothesis regarding the impact of democracy on military expenditures and government budgets, we suppose that significant shocks in political institutions influence the other variables in the model simultaneously. Military factors (military expenditures [% of GDP], arms imports [% of GDP], and armed forces personnel [% of total labor force]) and government budgets (non-military expenditures [% of GDP] and government revenue [% of GDP]) follow democracy factors (electoral democracy and military in politics indexes) in our Cholesky ordering.

Changes in political institutions affect government budgets by influencing expenditures (military and non-military) and government revenues. We employ an unrestricted panel-VAR model comprising seven variables to analyze the impact of democracy shocks on government expenditures and revenues in developing countries. The vector of endogenous variables in our PVAR model is given by:

$$Y_t = [elecdem,\ militpolit,\ milex,\ armimp,\ armforce,\ non\text{-}milex,\ govreven] \quad (2)$$

To avoid spurious regression results, it is essential to test the stationarity of the variables before performing the panel VAR analysis. Therefore, the series are tested to check for the existence of unit roots. Levin, Lin, and Chu (2002), Im, Pesaran, and Shin (2003), Maddala and Wu (1999), and Choi (2001) have offered different tests for unit roots in panel data. Levin, Lin, and Chu (2002) suggest a common unit root under the null hypothesis against the alternative of stationarity of all individuals, whereas the other tests assume individual unit roots under the null hypothesis. The results of the panel unit root test are presented in Table 1. We include a constant but not a time trend (Dickey and Fuller, 1979). The results in Table 1 reveal that all of the variables are stationary at their level based on the different unit root tests.[3] When the variables are stationary in levels, an unrestricted VAR model is employed.

Automatic lag length selection (Schwarz information criteria) is used. The null hypothesis for the Levin, Lin, and Chu test is a unit root that assumes a common unit root process. For the other three tests, the null hypothesis is a unit root that supposes an individual unit root process. ** indicates significance at the 5% level; * indicates significance at the 10% level.

To supplement the robustness of the unit root tests, we also carried out a so-called "second generation panel unit-root test' developed on the assumption of cross-sectional dependence among panel units.[4] Pesaran (2007) suggests a simple way of addressing cross-sectional dependence that does not require the estimation of factor loading. His method is based on augmenting the usual ADF regression with the lagged cross-sectional mean and its first difference to capture the cross-sectional dependence that arises through a single-factor model (Boubtane, Coulibaly, and Rault 2012). Table 2

Table 1. Panel unit root tests.

Variables	Levin, Lin & Chu	Im, Pesaran and Shin (IPS)	ADF-Fisher	PP-Fisher
armimp	−15.16[a]	−15.49[a]	436.2[a]	489.84[a]
milex	−7.84[a]	−6.85[a]	210.65[a]	230.26[a]
militpolit	−4.27[a]	−4.80[a]	151.1[a]	164.46[a]
elecdem	−2.69[a]	−3.11[a]	198.83[a]	238.83[a]
non-milex	−4.83[a]	−5.65[a]	201.82[a]	180.31[a]
govreven	−3.95[a]	−4.50[a]	171.92[a]	158.60[a]
armforce	−0.92	−1.57*	131.76[a]	129.97[a]

Table 2. CIPS unit root test.

Variables	CIPS (t-statistic)	Truncated CIPS (t-statistic)	*Critical values (5%)*	*Critical values (10%)*
armimp	−2.85[a]	−2.78[a]	*−2.13*	*−2.04*
milex	−2.53[a]	−2.52[a]	*−2.15*	*−2.05*
militpolit	NA	NA	-	-
elecdem	−2.05*	−2.05*	*−2.11*	*−2.04*
non-milex	−3.58[a]	−2.22[a]	*−2.20*	*−2.08*
govreven	−2.67[a]	−2.59[a]	*−2.20*	*−2.08*
armforce	−2.63[a]	−2.30[a]	*−2.11*	*−2.04*

[a]indicates significance at the 5% level; * indicates significance at the 10% level; Deterministic chosen: constant.

reports the results of the cross-sectionally augmented IPS ($CIPS = N^{-1}\sum_{i=1}^{N} CADF_i$), and the truncated version of the CIPS statistic where the individual cross-sectionally augmented Dickey-Fuller (CADF) statistics are suitably truncated to avoid the undue influences of extreme outcomes. Critical values reported in Pesaran (2007) are provided through Monte Carlo simulations for a specific specification of the deterministic component, and depend both on the cross-sectional and time series dimensions. The null hypothesis is the presence of a unit root. The results of Table 2 indicate that at the conventional levels of significance, our variables are stationary in levels.[5] Therefore, we will apply these variables in an unrestricted PVAR model and in their levels.

Impulse Responses to Electoral Democracy Index

Impulse response functions capture the effects of a one-time shock to electoral democracy on military factors, nonmilitary expenditures, and government revenues in the VAR model. If the disturbance terms ε_t are contemporaneously uncorrelated, interpreting the results of the impulse response functions is straightforward. The ith innovation ε_{it} is interpreted as a shock to the ith endogenous variableZ_{it}. The statistical significance of the impulse response is judged by the confidence bands around them (Runkle 1987). In this study, 68% confidence intervals for the IRFs are estimated (Sims and Zha 1999). In the estimated IRFs in Figure 1, the middle line displays the response of the variables to a one-standard-deviation positive shock in electoral democracy.[6] The dotted lines represent the confidence bands. The impulse responses are judged to be statistically insignificant wherever the horizontal line lies between two confidence bands (Berument, Ceylan, and Dogan 2010). The horizontal line in the IRFs displays the timeline, and the vertical line represents the magnitudes of the responses to shocks.

The results of the estimated VAR may also be sensitive to the number of lags selected for the variables in the model. Statistical criteria such as LR, final prediction error (FPE), Akaike information criterion (AIC), Schwarz information criterion (SC), and Hannan–Quinn information criterion (HQ) are usually applied to decide the optimal lag length in VAR models. As Table 3 shows, we select the lag length of one based on the SC criteria, as it is known to be more parsimonious (Pesaran and Smith 1998).[7]

Figure 1 shows the panel impulse responses of the democracy indexes, military factors, and government budgets to a one-standard-deviation positive shock in electoral democracy for the 1990–2017 period among the 47 developing countries. Electoral democracy is the first variable in our PVAR; it is followed by military in politics, military expenditures (% of GDP), arms imports (% of GDP), armed forces personnel (% of total labor force), non-military expenditures (% of GDP), and government revenues (% of GDP). We use the electoral component of democracy to mimic the effects of political institutions on military factors and government budgets in developing countries. Higher-quality political institutions will decrease military involvement in politics. This may influence military factors, including military expenditures, arms imports, and

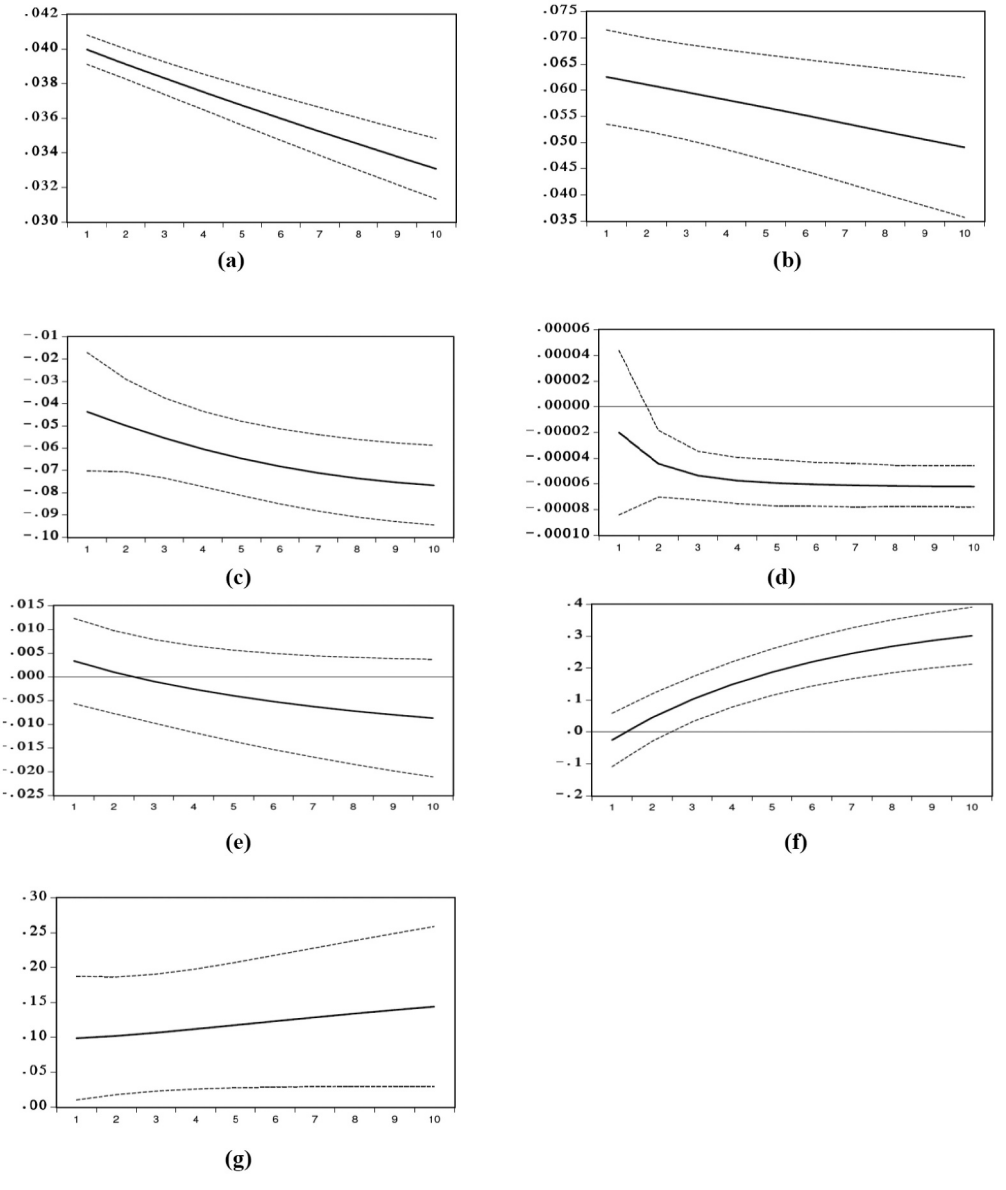


Figure 1. Impulse response functions related to a one-standard-deviation positive shock in electoral democracy index. **a** Response of electoral democracy index. **b** Response of military in politics index. **c** Response of military expenditures (% of GDP). **d** Response of arms imports (% of GDP). **e** Response of armed forces personnel (% of total labor force). **f** Response of non-military expenditure (% of GDP). **g** Response of government revenue (% of GDP). The dotted lines represent ±1 standard deviation. The horizontal axis shows the periods (years) after the initial shock. The magnitude of the responses is presented on the vertical axis.

armed forces personnel. Finally, changes in the political behavior of the government and its military expenditures may affect its spending on non-military sectors and improve government revenues. The results of Figure 1 show that the increases in the quality of electoral democracy have a statistically significant impact on the military in politics index (recall that higher values of this index mean less involvement of the military in politics). Improvement in political conditions has a negative and statistically significant impact on the ratios of military expenditures to GDP and arms imports to GDP (after one year). The responses of the ratio of armed

Table 3. Panel VAR lag order selection criteria.

Lag	LogL	LR	FPE	AIC	SC	HQ
0	−4999.04	-	0	14.92	14.96	14.93
1	1023.13	11,900.76	0	−2.88	−2.50[a]	−2.73
2	1118.94	187.34	0	−3.02	−2.31	−2.74[a]
3	1175.05	108.53	0[a]	−3.04[a]	2.00	−2.64
4	1216.60	79.50	0	−3.02	−1.65	−2.49

[a]indicates lag order selected by the criterion, LR: sequential modified LR test statistic (each test at 5% level); FPE, Final prediction error; AIC, Akaike information criterion; SC, Schwarz information criterion; HQ, Hannan–Quinn information criterion.

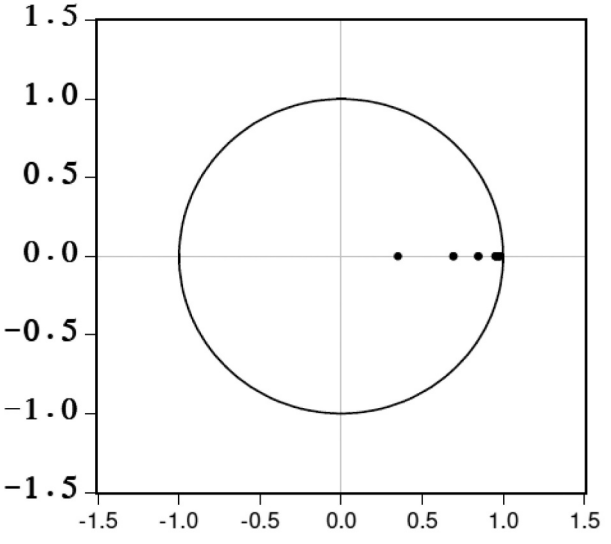


Figure 2. Inverse roots of AR characteristic polynomial.

forces personnel to total labor force (after one year) to the positive shocks in democracy are negative but statistically insignificant. Improvement in electoral democracy has a positive and statistically significant impact on the ratios of non-military expenditures to GDP (after one year) and government revenues to GDP. This indicates that a better quality of electoral democracy affects the government budget by creating a negative impact on the military sector and positive impact on the non-military sector and government revenues.

These results are in line with the previous findings of Lebovic (2001) who shows a positive association between the quality of democratic institutions and the share of non-military expenditures to the military expenditures. Our empirical results, however, are at odds with the theoretical discussion suggested by Besley and Robinson (2010) who discuss that democracies are able to spend more on military for national security purposes because they do not face that much of a threat of a coup from a strong military as dictatorships do. Nonetheless, developing countries with less developed political systems could be more susceptible to internal conflict and thus have more incentives to invest in their military (Besley and Persson 2009).

Figure 2 presents the AR graph reporting the inverse roots of the characteristic AR polynomial (Lütkepohl 1991). According to this figure, all roots in the PVAR model are located inside the unit circle and have a modulus of less than one. Thus, the estimated PVAR model is stable (stationary). The stability of estimated PVAR is important condition for validity of certain results such as impulse response standard errors.

Impulse Responses to Other Components of Democracy

The electoral element is an essential part of the V-Dem conceptual scheme. A democratic regime obviously has an electoral component. However, merely holding elections is not adequate. Countries may pretend to have 'democratic features' without being electorally democratic (Coppedge et al. 2015). Thus, as a robustness check, we consider the indexes of other components of democracy – liberal, participatory, deliberative, and egalitarian – that propose different approaches to defining democracy. These democracy indices range between 0 and 1, with higher values indicating a higher quality of democracy. Coppedge et al. (2015) describe these components as follows:

- The *liberal* component of democracy embodies the intrinsic value of protecting individual and minority rights against a potential 'tyranny of the majority'. This is achieved through constitutionally protected civil liberties, strong rule of law, and effective checks and balances that limit the use of executive power.
- The *participatory* component embodies the values of direct rule and active participation by citizens in all political processes; it emphasizes non-electoral forms of political participation, such as through civil society organizations and mechanisms of direct democracy.
- The *deliberative* component enshrines the core value that political decisions in pursuit of the public good should be informed by respectful and reasonable dialogue at all levels rather than by emotional appeals, solidary attachments, parochial interests, or coercion.
- The *egalitarian* component holds that material and immaterial inequalities inhibit the actual exercise of formal rights and liberties; hence, a more equal distribution of resources, education, and health across various groups should enhance political equality.

We use these components of democracy one by one in our panel VAR model to investigate the responses of our variable to the shocks in components of democracy. The overall results are largely consistent with our earlier findings, indicating that positive shocks to other aspects of democracy (liberal, participatory, deliberative, and egalitarian) reduce military involvement in politics, military expenditures (as % of GDP), and arms imports (as % of GDP). This creates an opportunity for developing countries to expand their non-military expenditures (as % of GDP) and increases government revenues (as % of GDP).[8]

Impulse Responses of Variables in per Capita Form to Polity2 Index

As a further robustness check, Figure 3 shows the impact of the Polity2 index of democratization on the military in politics index, military expenditures (per capita), arms imports (per capita), armed forces personnel (% of total labor force), non-military expenditures (per capita), and government revenues (per capita). The Polity2 indicator (*polity*) is an alternative measure of political systems, ranging from −10 (full autocracy) to 10 (full democracy). This index represents combinations of the autocratic and democratic characteristics of political institutions for various years (Marshall, Gurr, and Jaggers 2017).

In this model, considering the different lag criteria, we use two lags as the optimum number. The PVAR stability condition test (roots of characteristic polynomial) indicates that the PVAR satisfies the stability condition.[9] The positive shocks to the Polity2 index have a positive and statistically significant impact on the military in politics index (i.e. lower participation of military in politics) and a negative and statistically significant impact on per capita military expenditures (after two years) and per capita arms imports (after three years). The negative responses of the armed forces (% of total labor force) are only marginally significant eight years after the initial shock. The responses of per capita non-military expenditures to Polity2 shocks are positive and statistically significant after three years. We observe no significant responses of per capita government revenues to Polity2 shocks.

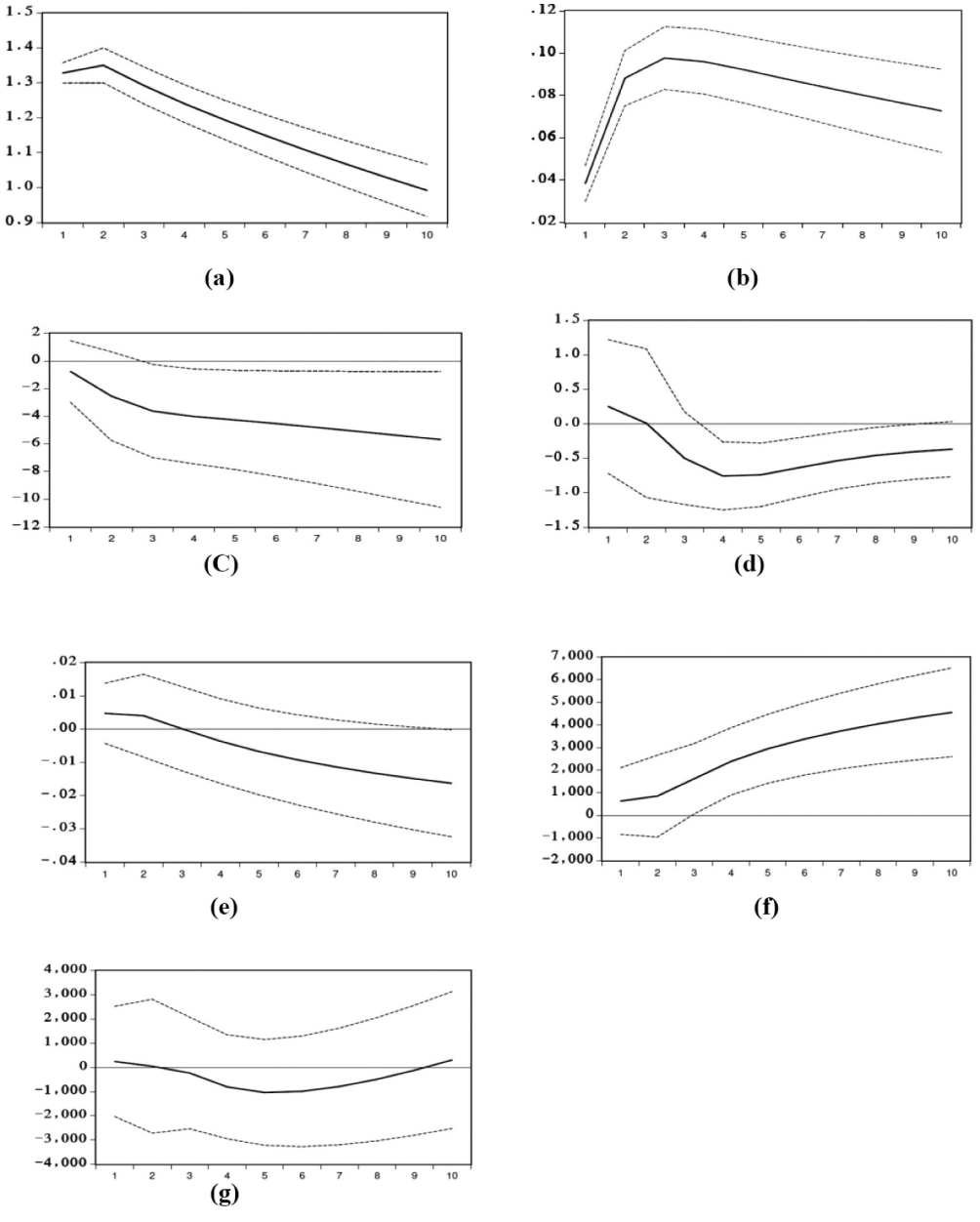


Figure 3. Impulse response functions related to a one-standard-deviation positive shock in Polity2 index. **a** Response of Polity2 index. **b** Response of military in politics index. **c** Response of military expenditures (per capita). **d** Response of arms imports (per capita). **e** Response of armed forces personnel (% of total labor force). **f** Response of non-military expenditure (per capita). **g** Response of government revenue (per capita). The dotted lines represent ±1 standard deviation. The horizontal axis shows the time periods (years) after the initial shock. The magnitude of the responses is presented on the vertical axis.

These results confirm our previous findings regarding the negative impacts of political institutions on military expenditures and arms imports and their positive impacts on non-military expenditures. Our empirical analysis is consistent with the theoretical model developed by Dizaji, Farzanegan, and Naghavi (2016) indicating that a country's political regime determines the structure of government spending to defend its assets, i.e. on the military, versus productivity enhancing non-military expenditure. They suggest that a democratic state, where more skilled workers play the role of the

median voter, prefers expenditure on labor productivity-enhancing public goods that expands its industry sector.

Impulse Responses of Democracy to Shocks in Military Expenditures

To examine our second hypothesis regarding the negative impact of military expenditures on democracy and non-military expenditures, we assume that significant shocks in military expenditures influence the other variables in the model simultaneously. Changes in military budgets affect arms imports and armed forces personnel, and thus military involvement in politics. Changes in military factors may affect the government's budget and political behavior. We employ an unrestricted PVAR model comprising seven variables to analyze the impact of military shocks on government budgets and political institutions in developing countries. The vector of endogenous variables in our PVAR model with focus on military shocks is given by:

$$Y_t = [\text{Milex, Armimp, Armforce, Militpolit, Non-milex, Govreven, Polity}] \quad (3)$$

Figure 4 shows the panel impulse responses of the democracy indexes, military factors, and government budgets to a one-standard-deviation positive shock in military expenditures (% of GDP) for the 1990–2017 period among the 47 developing countries. Military expenditure (% of GDP) is the first variable in our PVAR model, followed by arms imports (% of GDP), armed forces personnel (% of total labor force), military in politics, nonmilitary expenditures (% of GDP), government revenues (% of GDP), and the Polity 2 index. We use military expenditures (% of GDP) to mimic the effects of military expenditures on government budgets and political institutions in developing countries. Increases in military expenditures lead to positive changes in arms imports and armed forces personnel. These positive changes in military factors increase military involvement in politics. A government with more military characteristics will spend more on the military and less in non-military sectors. These factors have a negative impact on political institutions. The results of Figure 4 show that increases in military expenditures (% of GDP) have significant and positive impacts on arms imports (% of GDP) and armed forces personnel (after two years). The responses of the military in the politics index to military expenditures shock are negative and significant, indicating that increasing military expenditures encourages military involvement in politics. The responses of the ratios of non-military expenditures to GDP (after two years) and government revenues to GDP (before the third year) to positive changes in military expenditures (% of GDP) are negative and statistically significant. Finally, the response of the Polity2 index to increases in military expenditures (% of GDP) is negative and statistically significant (two years after the initial shock). This implies that militarization reduces the attention paid to non-military sectors and increases dictatorship in developing countries.[10]

Generalized Panel Impulse Responses of Democracy Indexes to Military Shocks

We seek to obtain more robust results regarding the impact of military shocks on aspects of democracy by defining a panel VAR model to capture the impacts of military expenditures (% of GDP) on the electoral, liberal, deliberative, egalitarian, and participatory components of the V-DEM democracy indexes. The ordering of these democracy indexes in the PVAR system is crucial, as different settings may result in different panel impulse response functions (Dizaji 2019b). Ideally, the theory should help us choose the best ordering so that changes in some variables will follow the changes of other variables rather than lead them. Wherever the economic theory is not clear about the appropriate ordering of variables, we must use some other logical setting for sensitivity analysis. The panel generalized impulse response function (PGIRF) analysis, which is based on Koop et al. (1996) and Pesaran and Smith (1998), provides a useful solution when the theory is not clear regarding the possible linkages between the variables. The PGIRFs offer an orthogonal set of

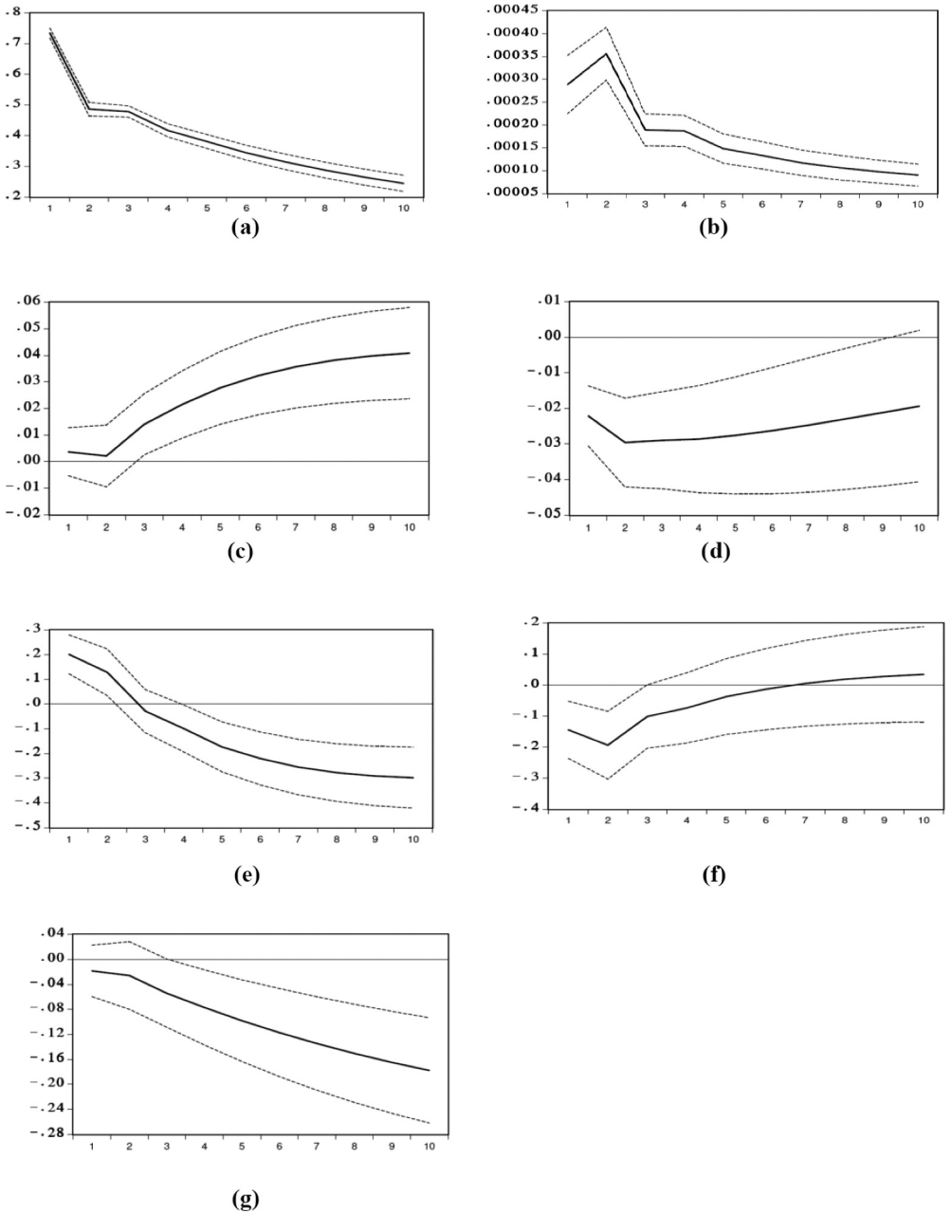


Figure 4. Impulse response functions related to a one-standard-deviation positive shock in military expenditures (% of GDP). **a** Response of military expenditures (% of GDP). **b** Response of arms imports (% of GDP). **c** Response of armed forces personnel (% of total labor force). **d** Response of military in politics index. **e** Response of non-military expenditure (% of GDP). **f** Response of government revenue (% of GDP). **g** Response of Polity2 index. The dotted lines represent ±1 standard deviation. The horizontal axis shows the time periods (years) after the initial shock. The magnitude of the responses is presented on the vertical axis.

innovations that are independent of the ordering of the variables in the PVAR model. Therefore, we estimate the panel generalized impulse responses of the democracy indexes (electoral, liberal, participatory, deliberative, and egalitarian) to a one-standard-deviation shock to military expenditures (as % of GDP).

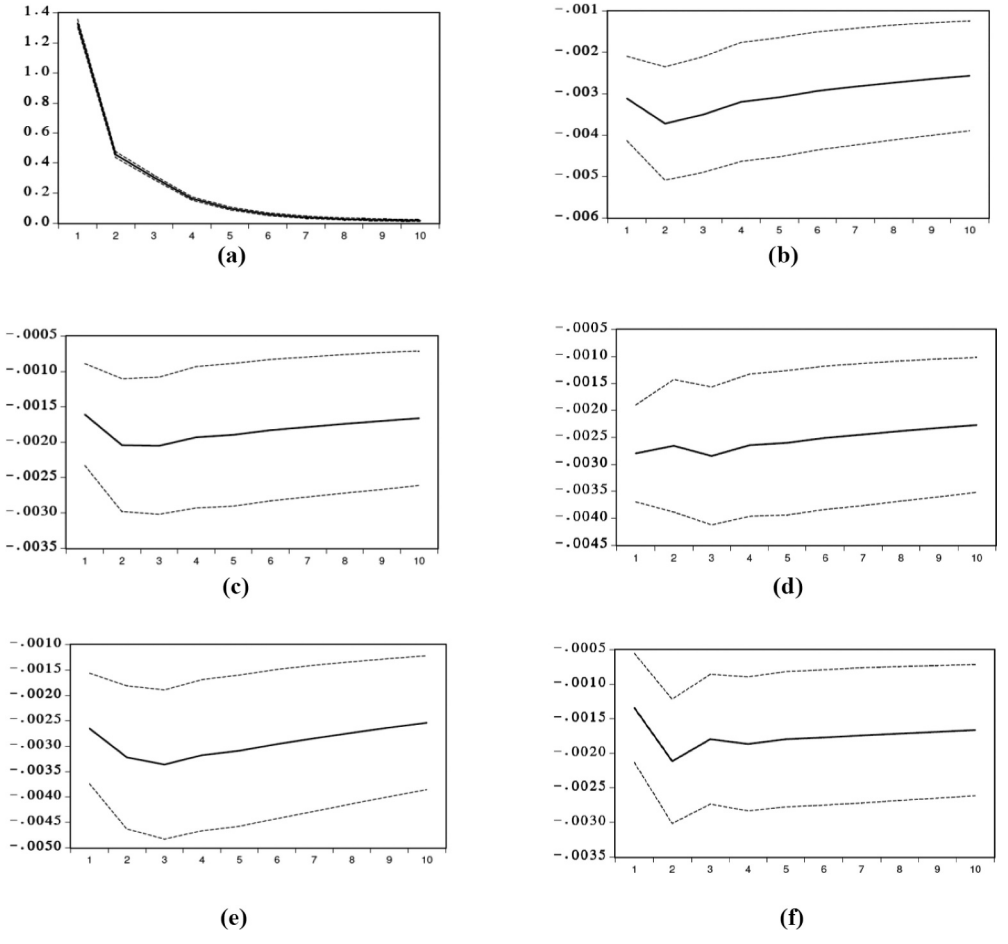


Figure 5. Impulse response functions related to generalized positive shock in military expenditures (% of GDP). **a** Response of military expenditures (% of GDP). **b** Response of electoral democracy. **c** Response of egalitarian democracy. **d** Response of liberal democracy. **e** Response of deliberative democracy. **f** Response of participatory democracy. The dotted lines represent ±1 standard deviation. The horizontal axis shows the time periods (years) after the initial shock. The magnitude of the responses is presented on the vertical axis.

As Figure 5 shows, the responses of the electoral democracy, egalitarian democracy, liberal democracy, deliberative democracy, and participatory democracy indexes during the entire period are negative and statistically significant to positive shocks in military expenditures (% of GDP).[11] The overall results indicate that militarization discourages electoral, liberal, deliberative, egalitarian, and participatory democracy in developing countries. In the majority of countries, the allocation of military expenditures is centralized, while non-military expenditures may be allocated at different layers of administration (e.g. by central, state or local governments). A higher share of military expenditures in an economy may increase the centralization of government budget allocation and the concentration of political power at the hands of political elites. A larger share of non-military expenditures may also be associated with more decentralized political power and the involvement of more middlemen (Teobaldelli 2011; Goel and Saunoris 2014).

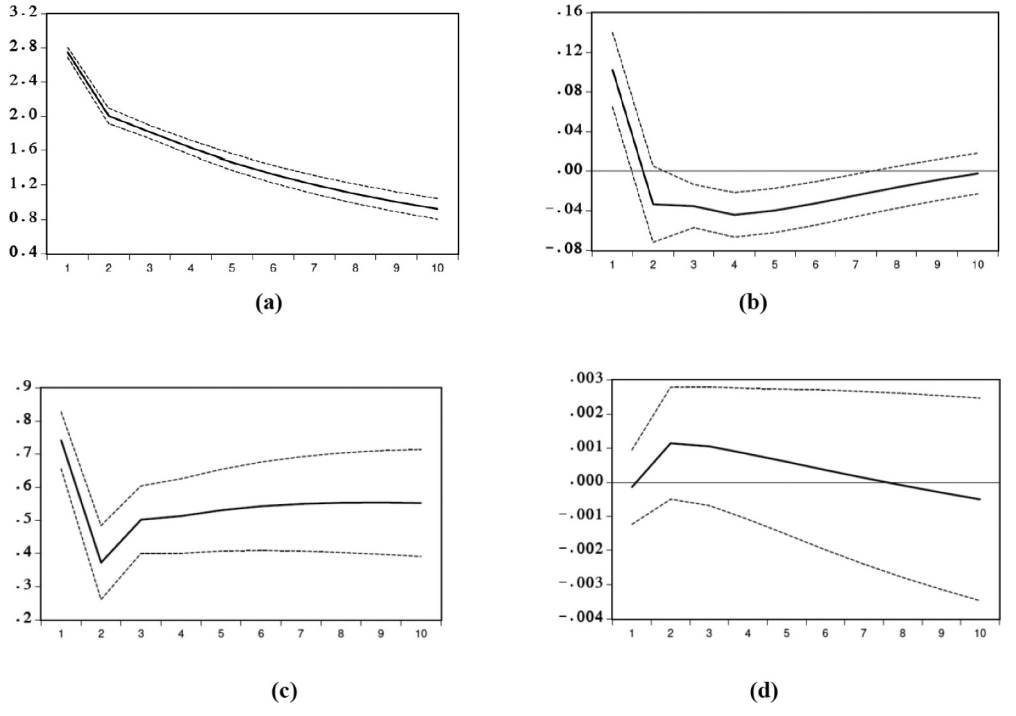


Figure 6. Impulse response functions related to a one-standard-deviation positive shock in non-military expenditures (% of GDP). **a** Response of non-military expenditures (% of GDP). **b** Response of military expenditures (% of GDP). **c** Response of government revenue (% of GDP). **d** Response of electoral democracy index. The dotted lines represent ±1 standard deviation. The horizontal axis shows the time periods (years) after the initial shock. The magnitude of the responses is presented on the vertical axis.

Impulse Responses to Shocks in Non-Military Expenditures

Figure 6 shows the panel impulse responses of non-military expenditures (% of GDP), military expenditures (% of GDP), government revenues (% of GDP), and electoral democracy to a one-standard-deviation positive shock in non-military expenditures (% of GDP) for the 1990–2017 period among the 47 developing countries.

As Figure 6 shows, the responses of military expenditures (% of GDP) to non-military expenditures (% of GDP) are negative and statistically significant (two years after the initial shock), whereas the responses of government revenues (% of GDP) are positive and statistically significant for the entire period. This suggests a substitution effect between military and non-military expenditures as percentages of GDP. Governments in developing countries can shift their expenditures from the military toward non-military sectors, which can increase government revenues (% of GDP). The political science literature has also shown evidence of a trade-off between public spending on military and social areas, such as education and health, and it suggests that this trade-off depends on the political regime (Deger 1985; Palmer 1990; Kimenyi and Mbaku 1995; Goldsmith 2003; Fordham and Walker 2005; Dizaji, Farzanegan, and Naghavi 2016).

The responses of the electoral democracy index to positive shocks in nonmilitary expenditures (% of GDP) are not significant, suggesting that the political behavior of governments in developing countries is influenced more by their spending on the military sector than by their spending on the nonmilitary sector.[12]

Variance Decomposition Analysis

Variance decomposition is a helpful tool for associating the variation of each variable with the shocks in all variables in the model. This informs us about the relative importance of each variable in causing the changes in a particular variable for different years after the shock. As Table 4 shows, the largest part of the changes in each variable is explained by its own shocks during the period, implying that the historical trend of each variable is important in explaining a large portion of its changes. The results of the variance decomposition analysis are in line with the results of the IRFs. This shows that the role of military expenditures (% of GDP) is more important than the role of non-military expenditures in determining the variations in the political variable. After its own changes, the largest part of the variations in the Polity2 index is explained by the changes in military expenditures (% of GDP) across all years. The role of military expenditures in explaining the changes in government revenues is negligible, while almost 12% of the variations in government revenues is explained by non-military expenditures after 30 years. These results indicate that government revenues are most influenced by non-military expenditures, while the militarization process is more important in affecting political institutions.

The shocks to government revenues (% of GDP) explain 37.83 % of the variations in non-military expenditures (% of GDP) and 11.77% of the variations in military expenditures (% of GDP) after 30 years. The role of the Polity2 index is also more important in explaining the changes in military expenditures (% of GDP) in the long run. The contribution of Polity2 shocks to military expenditure (as % of GDP) shocks was almost 1.9% in the tenth year, rising to approximately 9% in the thirtieth year, which is greater than its contribution in explaining non-military expenditures. This indicates that government revenues are more important in determining non-military expenditures, while the role of political institutions is more important in determining military expenditures.

Summary and Concluding Remarks

This study examines the response of political institutions to a positive shock in military expenditures and investigates the response of military expenditures to a positive shock in the quality of political institutions, while controlling for other factors. We use a panel of annual data from 47 developing countries covering 1990 to 2017 and apply a PVAR model to estimate the possible interrelationships among the variables. Our main results show that increases in the quality of the components of

Table 4. Variance decompositions of government expenditures/revenues and Polity2 index.

Year	Non-military expenditures (% GDP)	Military expenditures (% GDP)	Government revenues (% GDP)	Polity2
Non-military expenditures (% GDP)				
1	100	0	0	0
10	78.43	1.20	19.73	0.64
20	64.42	1.45	32.49	1.64
30	58.40	1.39	37.83	2.37
Military expenditures (% GDP)				
1	0.32	99.68	0	0
10	0.93	91.62	5.49	1.96
20	1.08	83.03	9.99	5.89
30	1.03	78.38	11.77	8.83
Government revenues (% GDP)				
1	7.36	0.09	92.55	0
10	8.59	0.06	91.35	0
20	10.86	0.07	89.08	0
30	11.85	0.06	88.08	0
Polity2				
1	0.06	0.08	0	99.86
10	0.52	1.75	0.61	97.11
20	0.93	3.27	1.62	94.17
30	1.10	4.07	2.47	92.36

democracy have statistically significant impacts on the military in politics index. Moreover, improvements in political institutions have a statistically significant negative impact on the ratios of military expenditures and arms imports to GDP. Furthermore, improvements in the democracy indexes have positive and statistically significant impacts on the ratios of non-military expenditures (e.g. education and health) and government revenues to GDP. This indicates that a better quality of democracy affects government budgets by exerting a negative impact on the military sector and positive impacts on the non-military sector and government revenues. We also investigate the response of democracy to positive shocks in military expenditures. Our results show that increases in military expenditures lead to positive changes in arms imports and armed forces personnel. These positive changes in military factors increase military involvement in politics. A government with more interaction with military interest groups will spend more on such fields and less in nonmilitary sectors. These factors create negative impacts on government revenues (as a percentage of GDP) and ultimately, on political institutions.

Improving the quality of democratic institutions in less developed countries has been one of the important foreign economic policies, particularly concerning policy makers and leaders from industrialized countries. Formulating such policies needs an understanding of the mechanism linking government fiscal policies with political institutions in emerging nations. While economists have mainly concentrated on the political logic of transitions to and from democracy (e.g. Wintrobe 1998; Dixit 2006), the military's role has received less attention. Our main results show that the response of democratic institutions to a positive shock in military expenditures burden is negative and significant in the global south countries, while its response to a shock in non-military expenditures are insignificant. This implies that the political behavior of governments in developing countries is influenced more by their spending on the military sector than by their spending on the nonmilitary sector. The results also suggest a substitution effect between military and nonmilitary expenditures as percentages of GDP.

One of the important implications of this research is that when trying to encourage or govern transitions to democracy, it is important that policy makers consider the complexities of the three-way relationships between political institutions, military expenditures and non-military expenditures in developing nations. Democracy would be highly achievable if authorities of the emerging countries adopt appropriate policies in order to decrease the share of military expenditures and arms in their budget. Consequently, the ratios of social expenditures and government revenues to GDP will also increase.

Finally, despite the aforementioned advantages of unrestricted PVAR model for the purposes of this study, we are aware of the intrinsic shortcomings of this approach in capturing the possible non-linear relationships among the variables.[13] Future research may employ other types of VAR models such as threshold VAR in order to examine the possible non-linear effects of government military or non-military expenditures on political institutions. Moreover, the final effect of military or non-military expenditures on the quality of political institutions may be contingent on other socio-demographic indicators of a particular country, which again is not possible to be captured with PVAR analysis. Therefore, and in the spirit of Rodrik (2007) and Dizaji, Farzanegan, and Naghavi (2016), future research may use case studies, which can reduce concerns about the important effect of heterogeneity between countries when examining the effect of government spending compositions on the quality of political institutions.

Notes

1. The relationship between corruption and military spending in the members of the Gulf Cooperation Council and other countries in the Middle East and North Africa region is also investigated by Farzanegan (2018).
2. www.imf.org/external/pubs/ft/weo/2019/02/weodata/index.aspx.
3. Armed forces personnel (as a percentage of the total labor force) is stationary with respect to Im, Pesaran and Shin, ADF-Fisher, and PP-Fisher tests.

4. To test for the presence of cross-sectional dependence in our data, we implemented the simple test of Pesaran (2004) and computed the cross-section Dependence (CD) statistic. This test is based on the average of pair-wise correlation coefficients of the ordinary-least-squares (OLS) residuals obtained from standard augmented Dickey-Fuller (ADF) regressions for each individual. Its null hypothesis is cross-sectional independence and is asymptotically distributed as a two-tailed standard normal distribution. The results (available upon request) indicate that the null hypothesis of cross-sectional independence for our variables is rejected.
5. The exception is for *militpolit*, which our statistical software does not provide evidence regarding its cross-section independence and stationarity. However, we have estimated our PVAR model and related impulse response functions with and without this variable. The main results do not change.
6. In general, the response of variable *j* to a unit shock (forecast error) in variable *k* is depicted graphically by IRFs to provide a visual impression of the dynamic interrelationships within the system. As economic variables and political indexes have different scales, we have considered innovations of one standard deviation rather than unit shocks in this study (Lütkepohl 2005). In this case, IRFs choose the units at the vertical axes equal to the standard deviations of the residuals corresponding to the variables whose effects are considered. Such a rescaling offers a better picture of the dynamic relationships because the average size of the innovations occurring in a system depends on their standard deviation.
7. We have also estimated our PVAR model by applying two lags (according to HQ) and three lags (according to FPE and AIC). The overall results do not change. These results are available upon request.
8. These results are available upon request.
9. This is available upon request.
10. Instead of using the Polity2 index of democratization in our panel VAR model, we also apply varieties of democracy indexes (i.e. electoral, liberal, deliberative, participatory, egalitarian democracy indexes) in separate panel VAR models one by one. The overall results confirm our earlier estimations, indicating that increased military expenditures have a negative impact on components of democracy in developing countries.
11. We also examine the generalized impulse responses of the electoral, liberal, deliberative, egalitarian, and participatory indexes to positive shocks in per capita military expenditures. The results reveal negative and statistically significant responses of the democracy indexes to positive shocks in per capita military expenditures. These results are available upon request.
12. We also apply other components of democracy (i.e. liberal, deliberative, egalitarian, participatory indexes) in separate panel VAR models. The results confirm our previous findings, showing that non-military spending (% of GDP) has a negative and significant impact on military spending (% of GDP) and a positive and statistically significant impact on government revenues (% of GDP). However, the responses of the democracy indexes to the shock in non-military expenditures (% of GDP) are not significant. These results are available upon request.
13. The possibility of a U-shaped relationship between democracy and public spending has been noted in some studies (Plümper and Martin 2003; Hausken, Martin, and Plümper 2004; Aidt, Dutta, and Dauntan 2010).

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Disclosure statement

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Ethical approval

This article does not contain any studies with human participants or animals performed by the authors.

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References

Aidt, T. S., J. Dutta, and M. Dauntan. 2010. "The Retrenchment Hypothesis and the Extension of the Franchise in England and Wales." *The Economic Journal* 120 (547): 990–1020. doi:10.1111/j.1468-0297.2009.02344.x.

Albalate, D., G. Bel, and F. Elias. 2012. "Institutional Determinants of Military Spending." *Journal of Comparative Economics* 40 (2): 279–290. doi:10.1016/j.jce.2011.12.006.

World Bank. 2019. *World Development Indicators*. World Bank,Washington D.C.

Bergougui, B., and M. Murshed. 2020. Revisiting the Oil and Democracy Nexus: New Evidence Utilizing V-DEM Democracy Data in a GMM PVAR Framework. ISS working paper, No. 656.

Berument, M. H., N. B. Ceylan, and N. Dogan. 2010. "The Impact of Oil Price Shocks on the Economic Growth of Selected MENA Countries." *Energy Journal* 31 (1): 149–176. doi:10.5547/0195-6574-EJ-Vol31-No1-7.

Besley, T., and J. A. Robinson. 2010. "Quis Custodiet Ipsos Custodes? Civilian Control over the Military." *Journal of the European Economic Association* 8 (2–3): 655–663. doi:10.1162/jeea.2010.8.2-3.655.

Besley, T., and T. Persson. 2009. "Repression or Civil War?" *American Economic Review* 99 (2): 292–297. doi:10.1257/aer.99.2.292.

Boubtane, E., D. Coulibaly, and C. Rault. 2012. Immigration, Growth and Unemployment: Panel VAR Evidence from OECD Countries, IZA working paper, No. 6966.

Bowman, K. S. 1996. "Taming the Tiger: Militarization and Democracy in Latin America." *Journal of Peace Research* 33 (3): 289–308. doi:10.1177/0022343396033003004.

Brauner, J. 2015. "Military Spending and Democracy." *Defence and Peace Economics* 26 (4): 409–423. doi:10.1080/10242694.2014.960245.

Brown, S., and M. Yücel. 1999. "Oil Prices and U.S. Aggregate Economic Activity: A Question of Neutrality." *Economic and Financial Policy Review* 11: 16–23.

Choi, I. 2001. "Unit root tests for panel data." Journal of International Money and Finance 20(2): 249–272. doi:10.1016/S0261-5606(00)00048-6

Collier, P., and A. Hoeffler. 2007. *Military Spending and the Risks of Coups D'etat. Centre for the Study of African Economies*. Department of Economics. Oxford University, Mimeo.

Coppedge, M., J. Gerring, S. I. Lindberg, J. Teorell, D. Altman, M. Bernhard, S. Fish, et al. 2015. *Varieties of Democracy: Codebook V4*. Varieties of Democracy (V-Dem) Project. University of Gothenborg, V-Dem Institute.

Dahlum, S., and C. H. Knutsen. 2017. "Do Democracies Provide Better Education? Revisiting the Democracy–human Capital Link." *World Development* 94: 186–199. doi:10.1016/j.worlddev.2017.01.001.

Dees, S., and J. Guntner. 2014. Analysing and Forecasting Price Dynamics across Euro Area Countries and Sectors: A Panel VAR Approach, European Central Bank Working Paper Series, No 1724.

Deger, S. 1985. "Human Resources, Government Education Expenditure and the Military Burden in Less Developed Countries." *Journal of Developing Areas* 20 (1): 37–48.

Dickey, D.A., and W.A. Fuller. 1979. "Distribution of the estimators for autoregressive time series with a unit root." Journal of the American Statistical Association 74(366): 427–431. doi:10.2307/2286348

Dixit, A. K., 2006. Predatory States and Failing States: An Agency Perspective, Princeton Research in Political Economy Working Paper.

Dizaji, S. F. 2014. "The Effects of Oil Shocks on Government Expenditures and Government Revenues Nexus (With an Application to Iran's Sanctions)." *Economic Modelling* 40: 299–313. doi:10.1016/j.econmod.2014.04.012.

Dizaji, S. F. 2019a. "Trade Openness, Political Institutions, and Military Spending (Evidence from Lifting Iran's Sanctions)." *Empirical Economics* 57 (6): 2013–2041. doi:10.1007/s00181-018-1528-2.

Dizaji, S. F. 2019b. The Potential Impact of Oil Sanctions on Military Spending and Democracy in the Middle East. ISS Working Paper, No. 644.

Dizaji, S. F., and M. R. Farzanegan. 2021. "Do Sanctions Constrain Military Spending of Iran?" *Defence and Peace Economics* 32 (2): 125–150. doi:10.1080/10242694.2019.1622059.

Dizaji, S. F., M. R. Farzanegan, and A. Naghavi. 2016. "Political Institutions and Government Spending Behavior: Theory and Evidence from Iran." *International Tax and Public Finance* 23 (3): 522–549. doi:10.1007/s10797-015-9378-8.

Dizaji, S. F., and S. M. Murshed. 2020. The Impact of External Arms Restrictions on Democracy and Conflict in Developing Countries, International Institute of Social Studies of Erasmus University Rotterdam, Working Paper, No. 661.

Dunne, P., and S. Perlo-Freeman. 2003. "The Demand for Military Spending in Developing Countries: A Dynamic Panel Analysis." *Defence and Peace Economics* 14 (6): 461–474. doi:10.1080/1024269032000085224.

Elbadawi, I. A., and P. Keefer. 2014. Democracy, Democratic Consolidation and Military Spending. *Working Papers 848*, Economic Research Forum, Cairo.

Enders, W. 1996. *RATS Handbook for Econometric Time Series*. New York: John Wiley & Sons.
Evans, G., and P. Rose. 2007. "Support for Democracy in Malawi: Does Schooling Matter?" *World Development* 33 (5): 904–919. doi:10.1016/j.worlddev.2006.09.011.
Farzanegan, M. R. 2011. "Oil Revenues Shocks and Government Spending Behavior in Iran." *Energy Economics* 33 (6): 1055–1069. doi:10.1016/j.eneco.2011.05.005.
Farzanegan, M. R. 2014. "Military Spending and Economic Growth: The Case of Iran." *Defence and Peace Economics* 25 (3): 247–269. doi:10.1080/10242694.2012.723160.
Farzanegan, M. R. 2018. "The Impact of Oil Rents on Military Spending in the GCC Region: Does Corruption Matter?" *Journal of Arabian Studies* 8 (1): 87–109. doi:10.1080/21534764.2018.1546938.
Farzanegan, M. R. 2021. "The Effects of International Sanctions on Iran's Military Spending: A Synthetic Control Analysis." *Defence and Peace Economics* 1–12. doi:10.1080/10242694.2021.1941548.
Farzanegan, M. R., and S. Witthuhn. 2017. "Corruption and Political Stability: Does the Youth Bulge Matter?" *European Journal of Political Economy* 49: 47–70. doi:10.1016/j.ejpoleco.2016.12.007.
Filippaki, A., and E. Mamatzakis. 2009. "Performance and Merton-type Default Risk of Listed Banks in the EU: A Panel VAR Approach." *Journal of Banking & Finance* 33 (11): 2050–2061. doi:10.1016/j.jbankfin.2009.05.009.
Fordham, B. O., and T. C. Walker. 2005. "Kantian Liberalism, Regime Type, and Military Resource Allocation: Do Democracies Spend Less?" *International Studies Quarterly* 49 (1): 141–157. doi:10.1111/j.0020-8833.2005.00338.x.
Goel, R. K., and J. W. Saunoris. 2014. "Military versus Non-military Government Spending and the Shadow Economy." *Economic Systems* 38 (3): 350–359. doi:10.1016/J.ECOSYS.2013.12.004.
Goldsmith, B. E. 2003. "Bearing the Defense Burden, 1886–1989: Why Spend More?" *Journal of Conflict Resolution* 47 (5): 551–573. doi:10.1177/0022002703254297.
Gupta, S., L. de Mello, and R. Sharan. 2001. "Corruption and Military Spending." *European Journal of Political Economy* 17 (4): 749–777. doi:10.1016/S0176-2680(01)00054-4.
Hausken, K., C. W. Martin, and T. Plümper. 2004. "Government Spending and Taxation in Democracies and Autocracies." *Constitutional Political Economy* 15 (3): 239–259. doi:10.1023/B:COPE.0000040431.47529.58.
Henderson, E., and D. Singer. 2000. "Civil War in the Post-colonial World, 1946–1992." *Journal of Peace Research* 37 (3): 275–299. doi:10.1177/0022343300037003001.
Im, K., M. Pesaran, and Y. Shin. 2003. "Testing for Unit Roots in Heterogeneous Panels." *Journal of Econometrics* 115 (1): 53–74. doi:10.1016/S0304-4076(03)00092-7.
Kimenyi, M., and J. Mbaku. 1996. "Rents, Military Elites, and Political Democracy." *European Journal of Political Economy* 11 (4): 699–708. doi:10.1016/0176-2680(95)00029-1.
Koop, G., M.H. Pesaran., and S.M. Potter. 1996. "Impulse response analysis in nonlinear multivariate models" Journal of Econometrics 115(1): 53–74. doi:10.1016/0304-4076(95)01753-4
Kotera, G., and K. Okada. 2017. "How Does Democratization Affect the Composition of Government Expenditure?" *Journal of Economic Behavior & Organization* 137 (C): 145–159. doi:10.1016/j.jebo.2017.03.004.
Lebovic, J. H. 2001. "Spending Priorities and Democratic Rule in Latin America." *Journal of Conflict Resolution* 45 (4): 427–452. doi:10.1177/0022002701045004002.
Levin, A., C.-F. Lin, and J. Chu. 2002. "Unit Root Tests in Panel Data: Asymptotic and Finite-sample Properties." *Journal of Econometrics* 108 (1): 1–24. doi:10.1016/S0304-4076(01)00098-7.
Love, I., and L. Zicchiano. 2006. "Financial Development and Dynamic Investment Behavior: Evidence from Panel VAR." *The Quarterly Journal Review of Economics and Finance* 46 (2): 190–210. doi:10.1016/j.qref.2005.11.007.
Lskavyan, V. 2011. "Democracy Levels and the Income-military Expenditure Relationship." *Applied Economics Letters* 18 (15): 1485–1489. doi:10.1080/13504851.2010.543067.
Lütkepohl, H. 1991. *Introduction to Multiple Time Series Analysis*. New York: Springer.
Lütkepohl, H. 2005. *New Introduction to Multiple Time Series Analysis*. Berlin: Springer-Verlag.
Maddala, G. S., and S. Wu. 1999. "A Comparative Study of Unit Root Tests with Panel Data and A New Simple Test." *Oxford Bulletin of Economics and Statistics* 61 (S1): 631–652. doi:10.1111/1468-0084.0610s1631.
Maizels, A., and M. K. Nissanke. 1986. "The Determinants of Military Expenditures in Developing Countries." *World Development* 14 (9): 1125–1140. doi:10.1016/0305-750X(86)90115-4.
Marshall, M. G., T. R. Gurr, and K. Jaggers. 2017. "*POLITY™ IV Project Political Regime Characteristics and Transitions, 1800-2015 Dataset Users' Manual*(1st ed.)." In *Center for Systemic Peace*. Vienna, VA.
Palmer, G. 1990. "Alliance Politics and Issue Areas: Determinants of Defense Spending." *American Journal of Political Science* 34 (1): 190–211. doi:10.2307/2111515.
Pereira, R. M. 2004. "Democracy and Military Expenditures: A Cross-country Analysis." *Brazilian Journal of Business Economics* 4 (1): 63–72.
Pesaran, M. H. 2004. "General Diagnostic Tests for Cross Section Dependence in Panels." Cambridge Working Papers in Economics no. 435 and CESifo Working Paper Series no. 1229, University of Cambridge, Cambridge, United Kingdom.
Pesaran, M. H. 2007. "A Simple Panel Unit Root Test in the Presence of Cross-section Dependence." *Journal of Applied Econometrics* 27 (2): 265–312. doi:10.1002/jae.951.

Pesaran, M. H., and R. Smith. 1998. "Structural Analysis of Cointegration VARs." *Journal of Economic Surveys* 12 (5): 471–505. doi:10.1111/1467-6419.00065.
Plümper, T., and C. W. Martin. 2003. "Democracy, Government Spending, and Economic Growth: A Political Economic Explanation of the Barro-effect." *Public Choice* 117 (1–2): 27–50. doi:10.1023/A:1026112530744.
Rodrik, D. 2007. *One Economics, Many Recipes: Globalization, Institutions, and Economic Growth*. Princeton, NJ: Princeton University Press.
Ross, M. L. 2008. "But Seriously: Does Oil Really Hinder Democracy?" UCLA Department of Political Science.
Rota, M. 2016. "Military Spending, Fiscal Capacity and the Democracy Puzzle." *Explorations in Economic History* 60 (C): 41–51. doi:10.1016/j.eeh.2015.11.002.
Rothschild, K. W. 1973. "Military Expenditure, Exports and Growth." *Kyklos* 26 (4): 808–814. doi:10.1111/j.1467-6435.1973.tb02777.x.
Runkle, D. E. 1987. "Vector Autoregression and Reality." *Journal of Business and Economic Statistics* 5 (4): 437–442. doi:10.2307/1391992.
Sims, C. A. 1980. "Macroeconomics and Reality." *Econometrica* 48 (1): 1–48. doi:10.2307/1912017.
Sims, C. A., and T. Zha. 1999. "Error Bands for Impulse Responses." *Econometrica* 67 (5): 1113–1156. doi:10.1111/1468-0262.00071.
SIPRI, 2020. Military Expenditure Database. Stockholm International Peace Research Institute (SIPRI): Stockholm. https://www.sipri.org/sites/default/files/2020-06/yb20_summary_en_v2.pdf
Stock, J.H., and M.W. Watson. 2001. "Vector Autoregressions." Journal of Economic Perspectives 15(4): 101–115. doi:10.1257/jep.15.4.101
Tang, R., and S. Tang. 2018. "Democracy's Unique Advantage in Promoting Economic Growth: Quantitative Evidence for a New Institutional Theory." *Kyklos* 71 (4): 642–666. doi:10.1111/kykl.12184.
Teobaldelli, D. 2011. "Federalism and the Shadow Economy." *Public Choice* 146 (3/4): 269–289. doi:10.1007/s11127-009-9590-0.
Töngür, Ü., and A. Y. Elveren. 2015. "Military Expenditures, Income Inequality, Welfare and Political Regimes: A Dynamic Panel Data Analysis." *Defence and Peace Economics* 26 (1): 49–74. doi:10.1080/10242694.2013.848577.
Wintrobe, R. 1998. *The Political Economy of Dictatorship*. Cambridge: Cambridge University Press.
Yildirim, J., and S. Sezgin. 2005. "Democracy and Military Expenditure: A Cross-country Evidence." *Transition Studies Review* 12 (1): 93–100. doi:10.1007/s11300-005-0037-0.

Appendix

List of countries

Algeria, Angola, Argentina, Bahrain, Bangladesh, Bolivia, Brazil, Chile, China, Colombia, Ecuador, Egypt, Ghana, Hungary, India, Indonesia, Iran, Jordan, Kazakhstan, Kuwait, Latvia, Lebanon, Lithuania, Malaysia, Mexico, Morocco, Myanmar, Nigeria, Oman, Pakistan, Peru, Philippines, Poland, Qatar, Romania, Saudi Arabia, South Africa, Sri Lanka, Sudan, Thailand, Tunisia, Turkey, UAE, Uruguay, Venezuela, Vietnam, Yemen.

Table A: Summary statistics

Variable	Mean	Standard deviation	Minimum	Maximum
armimp (arms imports as % of GDP)	0.2	0.3	0	4.1
milex (military expenditures as % of GDP)	3.125	4.215	0.276	117.34
non-milex (non-military expenditures as % of GDP)	25.296	9.01	4.386	86.82
armforce (armed forces personnel as % of total labor force)	1.954	1.882	0.075	12.848
govreven (government revenues as % of GDP)	25.919	10.756	3.797	72.34
militpolit (military in politics index)	3.377	1.556	0	6
polity (Polity2)	1.518	7.06	−10	10
elecdem (electoral democracy index)	0.447	0.274	0.014	0.929
libdem (liberal democracy index)	0.336	0.249	0.013	0.874
delibdem (deliberative democracy index)	0.353	0.243	0.016	0.879
egalitdem (egalitarian democracy index)	0.323	0.209	0.033	0.815
partibdem (participatory democracy index)	0.283	0.208	0.009	0.805